MD HUMAYUN

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Embedded & Systems Engineer | ESP32 | FreeRTOS | C++ | MQTT | GATE CS 93.8 Percentile

SUMMARY

M.Tech CS student at SVNIT Surat (GATE **93.8** percentile) building systems where the part that decides never waits on the part that reports: ESP32 firmware with safety logic and networking on separate cores, tested on the host in CI. Also built a multi-agent drone swarm simulator under a MeitY-sponsored project, with a paper under review at IEEE GSCON 2027, and an SDN DDoS mitigator measured under live attack traffic.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT Surat) *2025 – Present*
M.Tech in Computer Science (Information Security & Privacy) | ***CPI: 8.70/10*** **Surat, Gujarat**

Bihar Engineering University *2020 – 2024*
B.Tech in Computer Science Engineering | ***CGPA: 8.51/10*** **Patna, Bihar**

PROFESSIONAL EXPERIENCE

Research Intern — Hybrid AI-Based Autonomous Drone Swarm Navigation May 2026 – Jul 2026
MeitY (Ministry of Electronics & IT), Govt. of India — Sponsored Project, SVNIT Surat (Guide: Prof. Mukesh A. Zaveri)
Python / Blender / LLMs (Groq) / LoRA / PPO / Gymnasium *Team project (2)* | [Demo Video]

- Designed **BattleWorld**, a multi-agent simulator wrapped as a Gymnasium environment with a uniform **12-dimensional** state interface, so every commander — classical or learned — consumed identical input and the benchmark measured policy rather than input.
- Built the scenario pipeline: procedural urban scenes in Blender auto-labelled through an LLM annotation stage, making scenario count a parameter instead of a bottleneck.
- Fine-tuned an LLM with LoRA on the generated tactical labels, reaching **91.2%** mean token accuracy at **0.463** training loss.
- Benchmarked **seven** commander architectures over **28** held-out scenarios; hybrid LLM+PPO reached **67.9%** action accuracy against **60.7%** for the best classical baseline.

Salesforce Developer Intern Dec 2023 – Jan 2024
DST (Digital Skills Training) — Remote

- Developed Apex triggers and automated workflows in OOP-based backend logic, cutting manual data-entry overhead by **35%**.
- Customised KPI dashboards for **5+** cross-functional teams, replacing manually compiled weekly reports.

PUBLICATIONS

Hybrid AI-Based Autonomous Drone Swarm Navigation | *Guided by Prof. Mukesh A. Zaveri, SVNIT Surat* *Under Review, IEEE GSCON 2027 (#71435)*

PROJECTS

Smart Home Surveillance — ESP32 Alarm Firmware (v2.1) *2024 – 2026*
C++ / ESP32 / FreeRTOS / MQTT / Wokwi / GitHub Actions [GitHub]

- Split the firmware across the ESP32's two cores: the safety loop (sensor scan, arm/disarm state machine with **10 s** exit delay, **10 s** entry delay and **30 s** alarm, buzzer, LEDs, OLED) on core 1 with no blocking `delay()`, and Wi-Fi, NTP and MQTT as a FreeRTOS task on core 0, joined only by two queues, so a stalled network call cannot delay the siren.
- Reported state over MQTT with Last Will availability, retained state, sequence-numbered JSON events and a **16-deep** offline queue delivered in order after reconnect; remote DISARM is compiled out unless TLS and broker credentials are configured.
- Persisted the armed state in NVS across power cuts, supervised the loop with the task watchdog, debounced door and button inputs at **40 ms** and filtered the PIR with a **150 ms** confirmation window.
- Tested the state machine on the host in CI: **14** scenarios plus a TLS build check at three `millis()` start values, including 32-bit wrap-around, on arduino-esp32 2.x and 3.x (**84** runs, `-Wall -Wextra -Werror`).

IPS-ML / ShieldNet — Intrusion Prevention with Explainable AI (M.Tech Project) *Aug 2025 – Present*
Python / scikit-learn / XGBoost / SHAP / CICIDS2017 / pytest [GitHub]

- Trained six detectors and a soft-voting ensemble on **2,497,980** audited CICIDS2017 flows, evaluated on an identical held-out set of **749,394**; ensemble reached **0.8814** macro-F1 at a **0.00135** false alarm rate and **0.432 ms** median inference.
- Replaced argmax with a confidence-aware ALLOW / UNCERTAIN / BLOCK decision, choosing the operating point by sweeping the threshold against a stated **1%** false-alarm budget; routing **1.62%** of traffic to review cut missed attacks to **0.0149%** and benign disruption to **0.0486%**.
- Computed SHAP attributions inside the enforcement path and wrote them to an append-only audit log beside each action, so every block is explainable after the fact.

- Improved rare-class detection over the published Random Forest baseline without synthetic oversampling: SQL injection F1 **0.11** → **0.50**, Heartbleed **0.77** → **1.00**; covered by **29** unit tests with all thirteen plan deviations logged.

DDoS Detection & Mitigation — SDN + Machine Learning *Aug 2025 – Jan 2026*

Python | Mininet | POX | OpenFlow | Random Forest | Scapy [GitHub]

- Implemented detection inside a POX SDN controller tracking per-host packet rate, installing OpenFlow `DROP` rules in real time so mitigation is a control-plane action rather than an external appliance.
- Trained a Random Forest on SDN traffic features to replace the fixed-threshold baseline, which misfires on legitimate traffic spikes.
- Measured impact with `iperf` and `ping` rather than asserting it: throughput fell **54.3** → **31 Gbps** and average latency rose over **10x** under an ICMP flood.

Multi-Source Job Intelligence Platform *Apr 2026 – Present*

Python | Playwright | SQLite | GitHub Actions | Streamlit | Groq [GitHub]

- Built a headless scraping pipeline across **11** job boards and **9** government portals, persisting **2,158** listings to SQLite with idempotent upserts and per-scraper retry isolation, so one broken source degrades that source instead of failing the run.
- Applied fuzzy deduplication via `SequenceMatcher` at a tuned **0.88** threshold with weighted skill scoring, reducing **793** raw records to **357** ranked matches — a **55%** noise cut.
- Made LLM enrichment optional rather than required: a Groq-hosted model re-scores listings when configured, with a rule-based scorer as fallback, so the pipeline never depends on a paid service to run.
- Orchestrated an **8-stage** pipeline on GitHub Actions cron, fanning out to Excel workbooks, HTML digests, email alerts and a Streamlit dashboard with no manual trigger.

TECHNICAL SKILLS

Languages: C++ (primary), Python, MySQL, Bash
CS Core: Data Structures & Algorithms, System Design, OOP & Design Patterns, Operating Systems, DBMS, Computer Networks, Cryptography
ML / AI: TensorFlow, Keras, Scikit-learn, XGBoost, SHAP, LoRA, PPO, CNN, LSTM, VAE, Reinforcement Learning
LLM & Retrieval: LangChain, Pinecone, Ollama, Groq, RAG, Gymnasium
Embedded & IoT: ESP32, FreeRTOS, MQTT, Arduino, Raspberry Pi, I2C and GPIO, Wokwi simulation, host-side unit testing
Web / Tools: Flask (REST API), Docker, Git & GitHub, GitHub Actions, AWS (EC2, S3), Linux, SDN, OpenFlow, Mininet

CERTIFICATIONS & ACHIEVEMENTS

GATE CS 2025: 93.8 Percentile (National Eligibility for M.Tech Admissions)
NPTEL: Cloud Computing & Ethical Hacking — **All India Rank 33**
Summer Research Internship Certificate: SVNIT Surat, MeitY-Sponsored Project, issued by Prof. M. A. Zaveri
IIT Bombay: Python for Data Science, Computer Fundamentals | **IIT Roorkee:** IoT & Embedded Systems
Community: Founded NITIAN RAHI — engineering education community of **52,000+** across YouTube, Facebook and Instagram